

Construction of Equivalence

Zih-Yu Hsieh

August 3, 2026

Contents

1	Introduction (ND)	2
2	Preliminaries	3
2.1	Symmetric Monoidal Categories	3
2.2	Operads in Symmetric Monoidal Categories	3
2.3	Category $\mathcal{F}_G$	3
2.4	Equivariance Preliminaries	3
2.5	G -Categories	3
2.6	2-Categories	3
3	G-Operad $\mathcal{Q}_G$ in $\mathbf{Cat}_G$	4
3.1	The 2-Category $\mathcal{Q}_G\text{-PsAlg}$	4
4	2-Category $\mathcal{F}_G\text{-PsAlg}$	8
5	Construction of $\mathbb{R}_G$	9
6	Construction of $\mathbb{Q}_G$	10

1 Introduction (ND)

Fix G as a finite group. Let P_G be the *associativity G -operad* in $\mathbf{Set}_G$, defined by $P_G(n) := \mathbf{Set}(G, \Sigma_n)$, with structure maps defined by post composition

$$\gamma : \mathbf{Set}(G, \Sigma_k) \times \prod_{r=1}^k \mathbf{Set}(G, \Sigma_{j_r}) \cong \mathbf{Set} \left(G, \Sigma_k \times \prod_{r=1}^k \Sigma_{j_r} \right) \xrightarrow{\gamma \circ_-} \mathbf{Set}(G, \Sigma_j),$$

, where $j = \sum_{r=1}^k j_r$, and $\gamma : \Sigma_k \times \prod_{r=1}^k \Sigma_{j_r} \rightarrow \Sigma_j$ is the structure map for associativity operad $\mathcal{M}$ in $\mathbf{Set}$. This is naturally a free $(G \times \Sigma)$ -operad, as the right $(G \times \Sigma_n)$ -action at each $P_G(n)$ is free.

However, it is mentioned in [1] that such G -operad is not finitely generated, as a rough explanation, for every $n \in \mathbb{N}$, there exists $m > n$ and a function $f : G \rightarrow \Sigma_m$, such that it cannot be generated using $\{P_G(k)\}_{k \leq n}$ with the operadic structure map and the $(G \times \Sigma)$ -actions.

As a substitution, [1] defines a finitely-generated suboperad $Q_{G, \mathfrak{D}} \hookrightarrow P_G$ (for some parameter $\mathfrak{D}$ that's associated with choice of transitive G -sets), such that after chaotic generalization the two are equivalent in $\mathbf{Cat}_G$, and having equivalent categories of operadic algebras.

We'll mimic the statement, define an intermediate operad $Q_{G, \mathfrak{D}} \hookrightarrow Q_G \hookrightarrow P_G$, promote it to $\mathcal{Q}_G$ in $\mathbf{Cat}_G$, and modify the proof in [2] to prove the equivalence of 2-categories between $\mathcal{Q}_G\text{-PsAlg}$ and $\mathcal{F}_{G, \mathbb{R}}\text{-PsAlg}$.

Future work: Prove the equivalence of operads/algebras/pseudoalgebras with the original permutativity G -operad.

2 Preliminaries

2.1 Symmetric Monoidal Categories

2.2 Operads in Symmetric Monoidal Categories

2.3 Category $\mathcal{F}_G$

2.4 Equivariance Preliminaries

2.5 G -Categories

2.6 2-Categories

3 G -Operad $\mathcal{Q}_G$ in $\mathbf{Cat}_G$

Definition 3.1. Let Q_G be the suboperad of P_G , generated by all the group homomorphisms $\alpha : G \rightarrow \Sigma_n$ spanning through all $n \in \mathbb{N}$, the P_G structure maps γ_G , and the $(G \times \Sigma)$ -action on the elements. Define the operad $\mathcal{Q}_G := \mathcal{E}Q_G$ as the chaotic generalization in $\mathbf{Cat}_G$.

This operad is designed as a smallest suboperad that contains all group homomorphisms from G to a symmetric group, such that it's still a free $(G \times \Sigma)$ -operad.

Warning 3.2. If naively taken all group homomorphisms and their orbits under $(G \times \Sigma)$ -action, it doesn't form a suboperad in general for nontrivial group G . A counterexample can be constructed even for $G = \mathbb{Z}/2\mathbb{Z}$.

3.1 The 2-Category $\mathcal{Q}_G\text{-PsAlg}$

We mostly follow [2] in defining $\mathcal{Q}_G$ -pseudoalgebras.

Definition 3.3 ($\mathcal{Q}_G$ -pseudoalgebra). A $\mathcal{Q}_G$ -pseudoalgebra is a G -category $\mathcal{A}$ and operad action functors

$$\theta_j : \mathcal{Q}_G(j) \times \mathcal{A}^j \rightarrow \mathcal{A}$$

for each $n \in \mathbb{N}$, together with an invertible natural transformation ϕ_1 in the diagram

$$\begin{array}{ccc} \mathcal{A} & \xrightarrow{\iota} & \mathcal{Q}_G(1) \times \mathcal{A} \\ & \searrow \text{Id}_{\mathcal{A}} & \downarrow \theta_1 \\ & & \mathcal{A} \end{array} \quad \begin{array}{c} \swarrow \phi_1 \\ \end{array} \quad (3.4)$$

and a natural transformation $\phi = \phi(k; j_1, \dots, j_k)$ for each diagram

$$\begin{array}{ccc} \mathcal{Q}_G(k) \times \left(\prod_{r=1}^k \mathcal{Q}_G(j_r) \times \mathcal{A}^{j_r} \right) & \xrightarrow{\text{id} \times \prod_r \theta_{j_r}} & \mathcal{Q}_G(k) \times \mathcal{A}^k \\ \downarrow \text{shuffle} & \searrow \phi & \downarrow \theta_k \\ \mathcal{Q}_G(k) \times \prod_{r=1}^k \mathcal{Q}_G(j_r) \times \mathcal{A}^{j_r} & \xrightarrow{\gamma \times \text{id}} & \mathcal{Q}_G(j_+) \times \mathcal{A}^{j_+} \end{array} \quad \begin{array}{c} \uparrow \theta_{j_+} \\ \mathcal{A} \end{array} \quad (3.5)$$

such that the following strict equivariance hold for θ_j and ϕ :

(i) **(Equivariance of θ_j)** The following diagram commutes for $(g, \sigma) \in G \times \Sigma_j$:

$$\begin{array}{ccc} \mathcal{Q}_G(j) \times \mathcal{A}^j & \xrightarrow{(g, \sigma) \times (g, \sigma)^{-1}} & \mathcal{Q}_G(j) \times \mathcal{A}^j \\ & \searrow \theta_j & \swarrow \theta_j \\ & \mathcal{A} & \end{array} \quad (3.6)$$

(ii) **(Equivariance of ϕ)** The following equalities hold when given $g \in G$, $\sigma \in \Sigma_k$, and $\tau_r \in \Sigma_{j_r}$:

$$\phi(k; j_1, \dots, j_k) = \phi(k; j_{\sigma(1)}, \dots, j_{\sigma(k)}) \circ ((g, \sigma) \times (g, \sigma)^{-1})$$

$$\phi(k; j_1, \dots, j_k) = \phi(k; j_1, \dots, j_k) \circ (\text{id} \times \prod_r (\tau_r \times \tau_r^{-1}))$$

They also need to satisfy the following coherence properties:

- (iii) **(Operadic Composition)** The following pasting diagrams are equal, given $j_+ = \sum j_r$, $i_r = \sum_s i_{r,s}$, and $i_+ = \sum_{r,s} i_{r,s}$ with $1 \leq r \leq k$ and $1 \leq s \leq j_r$:

$$\begin{array}{ccccc}
\mathcal{Q}_G(k) \times \prod_r (\mathcal{Q}_G(j_r) \times \prod_s (\mathcal{Q}_G(i_{r,s}) \times \mathcal{A}^{i_{r,s}}))^{\text{id} \times \prod (\text{id} \times \theta_{i_{r,s}})} & \xrightarrow{\quad} & \mathcal{Q}_G(k) \times \prod_r (\mathcal{Q}_G(j_r) \times \mathcal{A}^{j_r}) & & \\
\downarrow \gamma \times \text{id} & & \downarrow \gamma & \searrow \text{id} \times \prod \theta_{j_r} & \\
\mathcal{Q}_G(j_+) \times \prod_{r,s} (\mathcal{Q}_G(i_{r,s}) \times \mathcal{A}^{i_{r,s}}) & \xrightarrow{\text{id} \times \prod \theta_{i_{r,s}}} & \mathcal{Q}_G(j_+) \times \mathcal{A}^{j_+} & \xleftarrow{\phi} & \mathcal{Q}_G(k) \times \mathcal{A}^k \\
& \searrow \gamma \times \text{id} & \downarrow \phi & \searrow \theta_{j_+} & \downarrow \theta_k \\
& & \mathcal{Q}_G(i_+) \times \mathcal{A}^{i_+} & \xrightarrow{\theta_{j_+}} & \mathcal{A}
\end{array}$$

equals

$$\begin{array}{ccccc}
\mathcal{Q}_G(k) \times \prod_r (\mathcal{Q}_G(j_r) \times \prod_s (\mathcal{Q}_G(i_{r,s}) \times \mathcal{A}^{i_{r,s}}))^{\text{id} \times \prod (\text{id} \times \theta_{i_{r,s}})} & \xrightarrow{\quad} & \mathcal{Q}_G(k) \times \prod_r (\mathcal{Q}_G(j_r) \times \mathcal{A}^{j_r}) & & \\
\downarrow \gamma \times \text{id} & \searrow \text{id} \times \prod (\gamma \times \text{id}) & \downarrow \text{id} \times \prod \phi & \searrow \text{id} \times \prod \theta_{j_r} & \\
\mathcal{Q}_G(j_+) \times \prod_{r,s} (\mathcal{Q}_G(i_{r,s}) \times \mathcal{A}^{i_{r,s}}) & & \mathcal{Q}_G(k) \times \prod_r (\mathcal{Q}_G(i_r) \times \mathcal{A}^{i_r}) & \xrightarrow{\text{id} \times \prod \gamma} & \mathcal{Q}_G(k) \times \mathcal{A}^k \\
& \searrow \gamma \times \text{id} & \downarrow \gamma \times \text{id} & \searrow \phi & \downarrow \theta_k \\
& & \mathcal{Q}_G(i_+) \times \mathcal{A}^{i_+} & \xrightarrow{\theta_{i_+}} & \mathcal{A}
\end{array}$$

(commuting squares are operadic associativity)

- (iv) **(Operadic Identity)** The following two pairs of two pasting diagrams are equal:

$$\begin{array}{ccccc}
\mathcal{Q}_G(k) \times \mathcal{A}^k & \xrightarrow{\text{id} \times \iota^k} & \mathcal{Q}_G(k) \times (\mathcal{Q}_G(1) \times \mathcal{A})^k & \xrightarrow{\text{id} \times \theta_1^k} & \mathcal{Q}_G(k) \times \mathcal{A}^k \\
& & \downarrow \text{shuffle} & & \\
& & \mathcal{Q}_G(k) \times \mathcal{Q}_G(1)^k \times \mathcal{A}^k & \searrow \phi & \downarrow \theta_k \\
& & \downarrow \gamma \times \text{id} & & \\
& & \mathcal{Q}_G(k) \times \mathcal{A}^k & \xrightarrow{\theta_k} & \mathcal{A}
\end{array}$$

equals

$$\begin{array}{ccccc}
& & \mathcal{Q}_G(k) \times (\mathcal{Q}_G(1) \times \mathcal{A})^k & & \\
& \nearrow \text{id} \times \iota^k & \downarrow \text{id} \times \Pi \phi & \nwarrow \text{id} \times \theta_1^k & \\
\mathcal{Q}_G(k) \times \mathcal{A}^k & \xrightarrow{\text{id}} & \mathcal{Q}_G(k) \times \mathcal{A}^k & \xrightarrow{\theta_k} & \mathcal{A} \\
\downarrow \text{id} \times \iota^k & \searrow \iota^k & \nearrow \gamma \times \text{id} & & \\
\mathcal{Q}_G(k) \times (\mathcal{Q}_G(1) \times \mathcal{A})^k & \xrightarrow{\text{shuffle}} & \mathcal{Q}_G(k) \times \mathcal{Q}_G(1)^k \times \mathcal{A}^k & &
\end{array}$$

(uses operadic associativity)

Also:

$$\begin{array}{ccccc}
\mathcal{Q}_G(k) \times \mathcal{A}^k & \xrightarrow{\iota} & \mathcal{Q}_G(1) \times \mathcal{Q}_G(k) \times \mathcal{A}^k & \xrightarrow{\text{id} \times \theta_k} & \mathcal{Q}_G(1) \times \mathcal{A} \\
& \searrow \text{id} & \downarrow \gamma \times \text{id} & \nearrow \phi & \downarrow \theta_1 \\
& & \mathcal{Q}_G(k) \times \mathcal{A}^k & \xrightarrow{\theta_k} & \mathcal{A}
\end{array}$$

equals

$$\begin{array}{ccc}
\mathcal{Q}_G(k) \times \mathcal{A}^k & \xrightarrow{\iota} & \mathcal{Q}_G(1) \times \mathcal{Q}_G(k) \times \mathcal{A}^k \\
\downarrow \theta_k & & \downarrow \text{id} \times \theta_k \\
\mathcal{A} & \xrightarrow{\iota} & \mathcal{Q}_G(1) \times \mathcal{A} \\
& \searrow \phi_1 & \downarrow \theta_1 \\
& & \mathcal{A}
\end{array}$$

(Uses operadic unitality)

Definition 3.7 ($\mathcal{Q}_G$ -pseudomorphism). A $\mathcal{Q}_G$ -pseudomorphism $(\mathbb{F}, \zeta) : (\mathcal{A}, \theta, \phi) \rightarrow (\mathcal{B}, \vartheta, \varphi)$ consists of G -functor $\mathbb{F} : \mathcal{A} \rightarrow \mathcal{B}$, such that for each j in the folloigin diagram commutes up to an invertibal natural transformation ζ_j :

$$\begin{array}{ccc}
\mathcal{Q}_G(j) \times \mathcal{A}^j & \xrightarrow{\text{id} \times \mathbb{F}^j} & \mathcal{Q}_G(j) \times \mathcal{B}^j \\
\downarrow \theta_j & \nearrow \zeta_j & \downarrow \vartheta_j \\
\mathcal{A} & \xrightarrow{\mathbb{F}} & \mathcal{B}
\end{array}$$

Moreover, they satisfy the following properties:

- (i) **(Equivariance of ζ)** For each $j, \zeta_j = \zeta_j \circ ((g, \sigma) \times (g, \sigma)^{-1})$ for all $(g, \sigma) \in G \times \Sigma_j$.
- (ii) **(Preserves Identity)** The following two pasting diagrams are equal:

$$\begin{array}{ccc}
\begin{array}{ccccc}
\mathcal{A} & \xrightarrow{\mathbb{F}} & \mathcal{B} & & \\
\downarrow \iota & & \downarrow \iota & \searrow \text{id} & \\
\mathcal{Q}_G(1) \times \mathcal{A} & \xrightarrow{\text{id} \times \mathbb{F}} & \mathcal{Q}_G(1) \times \mathcal{B} & \xrightarrow{\varphi_1} & \mathcal{B} \\
& \searrow \theta_1 & \uparrow \zeta_1^{-1} & \searrow \theta_1 & \\
& & \mathcal{A} & \xrightarrow{\mathbb{F}} & \mathcal{B}
\end{array} & = & \begin{array}{ccccc}
\mathcal{A} & \xrightarrow{\mathbb{F}} & \mathcal{B} & & \\
\downarrow \iota & \nearrow \text{id} & \downarrow \text{id} & & \\
\mathcal{Q}_G(1) \times \mathcal{A} & \xrightarrow{\theta_1} & \mathcal{A} & \xrightarrow{\mathbb{F}} & \mathcal{B}
\end{array}
\end{array}$$

(iii) **(Preserves Composition)** The following two pasting diagrams are equal:

$$\begin{array}{ccccc}
 \mathcal{Q}_G(k) \times \prod_r (\mathcal{Q}_G(j_r) \times \mathcal{A}^{j_r}) & \xrightarrow{\text{id} \times \prod (\text{id} \times \mathbb{F}^{j_r})} & \mathcal{Q}_G(k) \times \prod_r (\mathcal{Q}_G(j_r) \times \mathcal{B}^{j_r}) & & \\
 \downarrow \gamma & \searrow \text{id} \times \prod \theta_{j_r} & \swarrow \text{id} \times \prod \zeta_{j_r} & \searrow \text{id} \times \prod \vartheta_{j_r} & \\
 & \mathcal{Q}_G(k) \times \mathcal{A}^k & \xrightarrow{\text{id} \times \mathbb{F}^k} & \mathcal{Q}_G(k) \times \mathcal{B}^k & \\
 \swarrow \phi & \downarrow \theta_k & \searrow \zeta_k & \downarrow \vartheta_k & \\
 \mathcal{Q}_G(j) \times \mathcal{A}^j & \xrightarrow{\theta_j} & \mathcal{A} & \xrightarrow{\mathbb{F}} & \mathcal{B}
 \end{array}$$

equals

$$\begin{array}{ccccc}
 \mathcal{Q}_G(k) \times \prod_r (\mathcal{Q}_G(j_r) \times \mathcal{A}^{j_r}) & \xrightarrow{\text{id} \times \prod (\text{id} \times \mathbb{F}^{j_r})} & \mathcal{Q}_G(k) \times \prod_r (\mathcal{Q}_G(j_r) \times \mathcal{B}^{j_r}) & & \\
 \downarrow \gamma & & \downarrow \gamma & \searrow \text{id} \times \prod \theta_{j_r} & \\
 \mathcal{Q}_G(j) \times \mathcal{A}^j & \xrightarrow{\text{id} \times \mathbb{F}^j} & \mathcal{Q}_G(j) \times \mathcal{B}^j & \xrightarrow{\varphi} & \mathcal{Q}_G(k) \times \mathcal{B}^k \\
 \searrow \theta_j & \swarrow \zeta_j & \searrow \vartheta_j & \downarrow \vartheta_k & \\
 \mathcal{A} & \xrightarrow{\mathbb{F}} & \mathcal{B} & &
 \end{array}$$

Definition 3.8 ($\mathcal{Q}_G$ -2-cells). Let $(\mathbb{F}, \zeta), (\mathbb{G}, \xi) : (\mathcal{A}, \theta, \phi) \rightarrow (\mathcal{B}, \vartheta, \varphi)$ be two $\mathcal{Q}_G$ -pseudomorphisms, a $\mathcal{Q}_G$ -2-cell $\lambda : (\mathbb{F}, \zeta) \Rightarrow (\mathbb{G}, \xi)$ is a natural transformation $\lambda : \mathbb{F} \rightarrow \mathbb{G}$, such that the following two pasting diagrams are equal:

$$\begin{array}{ccc}
 \mathcal{Q}_G(j) \times \mathcal{A}^j & \xrightarrow{\text{id} \times \mathbb{F}^j} & \mathcal{Q}_G(j) \times \mathcal{B}^j \\
 \downarrow \theta_j & \searrow \text{id} \times \mathbb{G}^j & \downarrow \vartheta_j \\
 \mathcal{A} & \xrightarrow{\mathbb{G}} & \mathcal{B}
 \end{array}
 \quad \text{with } \xi_j \text{ as a 2-cell from } \text{id} \times \mathbb{G}^j \text{ to } \text{id} \times \mathbb{F}^j$$

$$\begin{array}{ccc}
 \mathcal{Q}_G(j) \times \mathcal{A}^j & \xrightarrow{\text{id} \times \mathbb{F}^j} & \mathcal{Q}_G(j) \times \mathcal{B}^j \\
 \downarrow \theta_j & \searrow \zeta_j & \downarrow \vartheta_j \\
 \mathcal{A} & \xrightarrow{\mathbb{F}} & \mathcal{B} \\
 & \searrow \lambda & \\
 & \mathcal{B} &
 \end{array}
 \quad \text{with } \lambda \text{ as a 2-cell from } \mathbb{F} \text{ to } \mathbb{G}$$

4 2-Category $\mathcal{F}_G\text{-PsAlg}$

In the category $\mathcal{F}_G$ (all based finite G -sets), the homsets $\mathcal{F}_G(\mathbf{n}^\alpha, \mathbf{m}^\beta)$ can also be viewed as a discrete category. We'll use this identification to define some notions.

Definition 4.1 ($\mathcal{F}_G$ -pseudoalgebra). An $\mathcal{F}_G$ -pseudoalgebra, is a pseudofunctor $\tilde{\mathcal{A}} : \mathcal{F}_G \rightarrow \mathbf{Cat}_G$, such that restricting to the full subcategory $\Pi_G \subset \mathcal{F}_G$, $\tilde{\mathcal{A}}$ is a strict functor. Moreover, we require it to be reduced, namely $\tilde{\mathcal{A}}(0) = *$ (the terminal G -category), and each hom-functor

$$\tilde{\mathcal{A}}_{\mathbf{n}^\alpha, \mathbf{m}^\beta} : \mathcal{F}_G(\mathbf{n}^\alpha, \mathbf{m}^\beta) \rightarrow \mathbf{Cat}_G(\tilde{\mathcal{A}}(\mathbf{n}^\alpha), \tilde{\mathcal{A}}(\mathbf{m}^\beta))$$

is a G -functor.

Question: In operad case, we have natural transformation between any two G -actions, do we need it here?

Let $\delta_i : \mathbf{n}^\alpha \rightarrow \mathbf{1}$ be the i th projection (namely $i \mapsto 1$, and any $i \neq j \mapsto 0$), since it belongs to Π_G , it generates a functor $\tilde{\mathcal{A}}\delta_i : \tilde{\mathcal{A}}(\mathbf{n}^\alpha) \rightarrow \tilde{\mathcal{A}}(\mathbf{1})$. Hence, by universality of product, all $\tilde{\mathcal{A}}\delta_i$ ($1 \leq i \leq n$) induces a functor $\delta : \tilde{\mathcal{A}}(\mathbf{n}^\alpha) \rightarrow \tilde{\mathcal{A}}(\mathbf{1})^n$.

Definition 4.2 ($\mathcal{F}_{G, \text{vs}}$ and $\mathcal{F}_{G, \mathbb{R}}$ -pseudoalgebra). An $\mathcal{F}_G$ -pseudoalgebra $\tilde{\mathcal{A}}$ is *very special* (or an $\mathcal{F}_{G, \text{vs}}$ -pseudoalgebra), if all $\delta : \tilde{\mathcal{A}}(\mathbf{n}^\alpha) \rightarrow \tilde{\mathcal{A}}(\mathbf{1})^{\mathbf{n}^\alpha}$ is an isomorphism for all $\mathbf{n}^\alpha \in \mathcal{F}_G$, and $\tilde{\mathcal{A}}$ is *strictly special* (or an $\mathcal{F}_{G, \mathbb{R}}$ -pseudoalgebra), if all δ is in fact identity.

Definition 4.3 ($\mathcal{F}_G$ -1-cell). Given $\tilde{A}, \tilde{B}$ two $\mathcal{F}_G$ -pseudoalgebras, a morphism (or $\mathcal{F}_G$ -1-cell) between them is a pseudo-transformation $(\mathbb{F}, \zeta) : \tilde{\mathcal{A}} \rightarrow \tilde{\mathcal{B}}$, such that restricting to Π_G , it becomes an actual natural transformation.

Fill in details for the commutativity diagrams

Definition 4.4 ($\mathcal{F}_G$ -2-cell). Given $\tilde{A}, \tilde{B}$ two $\mathcal{F}_G$ -pseudoalgebras, and two $\mathcal{F}_G$ -1-cells $(\mathbb{F}, \zeta), (\mathbb{G}, \xi) : \tilde{A} \rightarrow \tilde{B}$, an $\mathcal{F}_G$ -2-cell is a transformation $\lambda : \mathbb{F} \Rightarrow \mathbb{G}$, such that for any morphism $f : \mathbf{n}^\alpha \rightarrow \mathbf{m}^\beta$ in $\mathcal{F}_G$, the following two pasting diagrams are equal:

$$\begin{array}{ccc} \begin{array}{ccc} & \xrightarrow{\mathbb{F}_{\mathbf{n}^\alpha}} & \\ \tilde{\mathcal{A}}(\mathbf{n}^\alpha) & \xrightarrow{\lambda} & \tilde{\mathcal{B}}(\mathbf{n}^\alpha) \\ \tilde{\mathcal{A}}f \downarrow & \swarrow \mathbb{G}_{\mathbf{n}^\alpha} \quad \searrow \xi f & \downarrow \tilde{\mathcal{B}}f \\ \tilde{A}(\mathbf{m}^\beta) & \xrightarrow{\mathbb{G}_{\mathbf{m}^\beta}} & \tilde{B}(\mathbf{m}^\beta) \end{array} & = & \begin{array}{ccc} & \xrightarrow{\mathbb{F}_{\mathbf{n}^\alpha}} & \\ \tilde{\mathcal{A}}(\mathbf{n}^\alpha) & \xrightarrow{\mathbb{F}_{\mathbf{n}^\alpha}} & \tilde{\mathcal{B}}(\mathbf{n}^\alpha) \\ \tilde{\mathcal{A}}f \downarrow & \swarrow \zeta f \quad \searrow \mathbb{F}_{\mathbf{m}^\beta} & \downarrow \tilde{\mathcal{B}}f \\ \tilde{A}(\mathbf{m}^\beta) & \xrightarrow{\lambda} & \tilde{B}(\mathbf{m}^\beta) \\ & \searrow \mathbb{G}_{\mathbf{m}^\beta} & \end{array} \end{array}$$

5 Construction of $\mathbb{R}_G$

6 Construction of $\mathbb{Q}_G$

References

- [1] Kayleigh Bangs, Skye Binegar, Young Kim, Kyle Ormsby, Angélica M. Osorno, David Tamas-Parris and Livia Xu. Biased Permutative Equivariant Categories. *Homology, Homotopy and Applications*, vol. 23. 2021. <https://arxiv.org/abs/1907.00933>
- [2] Jiasen Liu, J. P. May, Kyle I. Roke, Hongyi Zhang and Keming Zhou. What are Symmetric Monoidal Categories? Available as ArXiv:2607.13912, 2026.